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|---|-----|------|---|----------|-----|
| 6 | (a) | (i) | <i>either</i> prevent loss of magnetic flux
<i>or</i> improves flux linkage with secondary | B1 | [1] |
| | | (ii) | <u>reduces</u> eddy current (losses)
<u>reduces</u> losses of energy (in core) | B1
B1 | [2] |
| | (b) | (i) | (induced) e.m.f. proportional to / equal to
rate of change of (magnetic) flux (linkage) | M1
A1 | [2] |
| | | (ii) | changing current in primary gives rise to | (1) | |
| | | | changing flux in core | (1) | |
| | | | flux links with the secondary coil | (1) | |
| | | | changing flux in secondary coil, inducing e.m.f. | (1) | |

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- (any three, 1 each to max 3) B3 [3]
- (c) e.g. can change voltage easily / efficiently
 high voltage transmission reduces power losses
- (any two sensible suggestions, 1 each) B2 [2]

3 (a) e.g. 'incandescent' emission (of electrons)